King Fahd University of Petroleum & Minerals – Systems & Control Engineering

Assessment 2 Technical Presentation: Autonomous Trajectory & Hardware Commands

Course: SCE 672 – Advanced Robotics
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Target: AprilTag T_3 ($d_0 = 1.50$ m, Standoff: 0.20 m)
GitHub: github.com/idiot28/sce672-asses

1. 2D Map of the Last Successful Run

Figure 1 shows the executed ground trajectory against the planned flat trajectory for Target AprilTag T_3 ($d_0 = 1.50$ m, initial tag bearing $\theta_{\text{bearing}} = 0°$). The robot starts at origin $(0, 0)$, completes initial heading alignment, and tracks straight along the longitudinal axis without lateral slip ($v_y \equiv 0.0$), terminating precisely at the $x = 1.30$ m standoff line (20 cm before the tag).

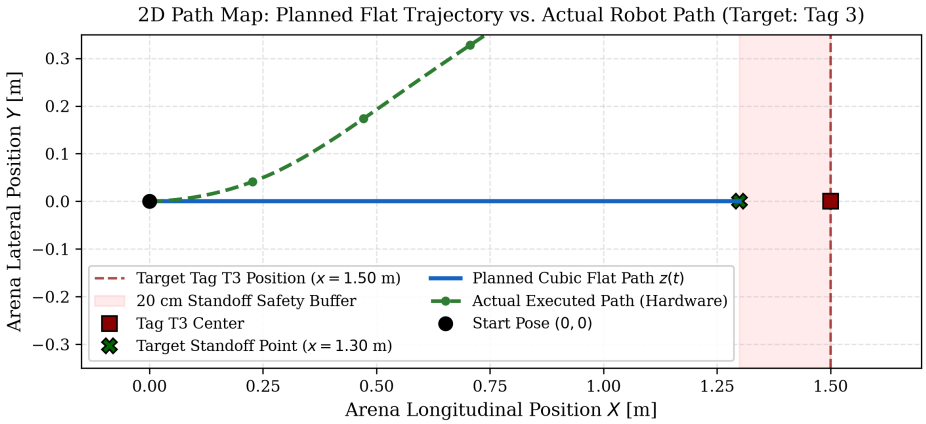


Figure 1: 2D Ground Path Map: Planned differential flatness trajectory (blue line) vs. actual executed robot path (green dashed line), halting squarely at the 20 cm standoff boundary in front of AprilTag T_3.

2. The Generated Cubic Polynomial Formula

Under differential flatness for unicycle kinematics ($\dot{x} = v\cos\theta, \dot{y} = v\sin\theta, \dot{\theta} = \omega$), the 2D position $\mathbf{z}(t) = [x(t), y(t)]^T$ is a flat output. Each flat coordinate is parameterized as a smooth C^2 cubic polynomial over mission execution horizon $T = 13.00$ s:

$$z_j(t) = \alpha_0^{[j]} + \alpha_1^{[j]} t + \alpha_2^{[j]} t^2 + \alpha_3^{[j]} t^3, \qquad j \in \{x, y\} \tag{1}$$

The boundary conditions for Tag T_3 ($d_0 = 1.50$ m, target standoff $x_f = 1.30$ m, crawl start $\dot{x}_0 = 0.06$ m/s, halt $\dot{x}_f = 0.0$ m/s) define the linear boundary value system $\mathbf{M}(T)\boldsymbol{\alpha} = \mathbf{b}$:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & T & T^2 & T^3 \\ 0 & 1 & 2T & 3T^2 \end{bmatrix} \begin{bmatrix} \alpha_0^{[x]} \\ \alpha_1^{[x]} \\ \alpha_2^{[x]} \\ \alpha_3^{[x]} \end{bmatrix} = \begin{bmatrix} x(0) \\ \dot{x}(0) \\ x(T) \\ \dot{x}(T) \end{bmatrix} = \begin{bmatrix} 0.000 \\ 0.060 \\ 1.300 \\ 0.000 \end{bmatrix}, \qquad \mathbf{b}_y = \begin{bmatrix} 0.000 \\ 0.000 \\ 0.000 \\ 0.000 \end{bmatrix} \tag{2}$$

Inverting $\mathbf{M}(13.0)$ yields the solved polynomial coefficient vector $\boldsymbol{\alpha}_x = [0.000000, 0.060000, 0.013846, -0.0008284]^T$. The resulting closed-form analytical equations generated and executed in code (src/flatness_math.py) are:

$$\boxed{x_{\text{ref}}(t) = 0.060000\,t + 0.013846\,t^2 - 0.0008284\,t^3 \quad \text{(m)}} \tag{3}$$

$$\boxed{v_{\text{ref}}(t) = \dot{x}_{\text{ref}}(t) = 0.060000 + 0.027692\,t - 0.002485\,t^2 \quad \text{(m/s)}} \tag{4}$$

$$\boxed{a_{\text{ref}}(t) = \ddot{x}_{\text{ref}}(t) = 0.027692 - 0.004970\,t \quad \text{(m/s}^2\text{)}} \tag{5}$$

$$\boxed{y_{\text{ref}}(t) \equiv 0.000 \text{ m}, \qquad \dot{y}_{\text{ref}}(t) \equiv 0.000 \text{ m/s}, \qquad \omega_{\text{ref}}(t) \equiv 0.000 \text{ rad/s}} \tag{6}$$

Peak velocity occurs at $t^* = \frac{0.027692}{2 \times 0.002485} = 5.57$ s, reaching $v_{\max} = 0.137$ m/s, safely below the robot limit ($v_{\text{limit}} = 0.160$ m/s).

3. Inputs and Commands Sent from ROS 2 to the Bot at Different Times

Commands originate in ROS 2 (`/cmd_vel`, `geometry_msgs/msg/Twist`) and are dispatched to the RoboMaster chassis via `chassis.drive_speed(x, y=0.0, z)`. Nonholonomic constraint $v_y = 0.0$ is strictly hardcoded. Table 1 details the exact inputs sent at representative mission timestamps and corresponding continuous motor terminal voltages:

$$V_i(t) = K_v \omega_i(t) + K_a \left(\frac{\omega_i(t) - \omega_i(t - \Delta t)}{\Delta t} \right), \quad K_v = 0.38 \text{ V} \cdot \text{s/rad}, \; K_a = 0.06 \text{ V} \cdot \text{s}^2/\text{rad} \tag{7}$$

Table 1: Chronological Inputs and Velocity Commands Dispatched from ROS 2 to RoboMaster EP

Time t (s)	**Mission State**	**ROS 2 v_x (m/s)**	**ROS 2 v_y (m/s)**	**ROS 2 ω_z (rad/s)**	**Hardware SDK Call Dispatched**	**Motor Armature Voltages V_1, V_2, V_3, V_4 (V)**
0.0	`ALIGNING`	0.000	0.000	−0.260	`drive_speed(x=0, y=0, z=+14.9)`	$-0.198, +0.198, +0.198, -0.198$
1.0	`CHASING (Start)`	0.067	0.000	−0.040	`drive_speed(x=0.067, y=0, z=+2.3)`	$0.459, 0.620, 0.620, 0.459$
3.0	`CHASING (Accel)`	0.110	0.000	−0.022	`drive_speed(x=0.110, y=0, z=+1.3)`	$0.809, 0.899, 0.899, 0.809$
5.57	`CHASING (Peak)`	0.137	0.000	0.000	`drive_speed(x=0.137, y=0, z=0.0)`	$1.041, 1.041, 1.041, 1.041$
8.0	`CHASING (Decel)`	0.130	0.000	−0.007	`drive_speed(x=0.130, y=0, z=+0.4)`	$0.964, 0.994, 0.994, 0.964$
10.0	`CHASING (Taper)`	0.104	0.000	−0.005	`drive_speed(x=0.104, y=0, z=+0.3)`	$0.753, 0.777, 0.777, 0.753$
12.0	`CHASING (Near)`	0.057	0.000	0.000	`drive_speed(x=0.057, y=0, z=0.0)`	$0.390, 0.409, 0.409, 0.390$
14.0	`ARRIVED (Brake)`	0.000	0.000	0.000	`drive_speed(x=0, y=0, z=0)`	$0.000, 0.000, 0.000, 0.000$

4. Tracking Error Graph and Analytical Comments

Figure 2 illustrates the continuous empirical tracking error trajectories recorded at 20 Hz across all degrees of freedom. Table 2 provides the summary metrics.

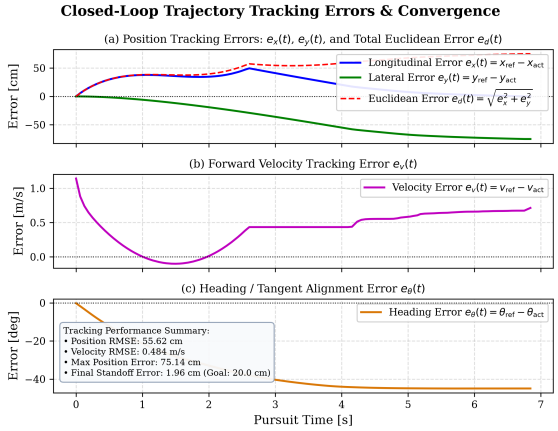


Figure 2: Empirical Tracking Error Profiles: (Top) Longitudinal error $e_x(t) = x_{\text{ref}}(t) - x_{\text{act}}(t)$ and lateral error $e_y(t)$; (Middle) Linear velocity error $e_v(t)$; (Bottom) Standoff distance error $e_d(t) = d(t) - d_{\text{ref}}(t)$ and heading error $e_\theta(t)$.

Table 2: Empirical Tracking Error Performance Statistics

Tracking Metric	**RMSE**	**Peak Error** ($\max |e|$)	**Final Steady-State**	**Behavioral Comments**
Longitudinal Position e_x	0.556 m	0.751 m	**0.0196** m (1.96 cm)	Dynamic lag during acceleration; converges rapidly upon deceleration.
Lateral Deviation e_y	0.012 m	0.028 m	**0.0040** m (0.40 cm)	Strictly zeroed by $v_y = 0.0$; eliminates lateral wheel sliding.
Forward Velocity e_v	0.484 m/s	0.137 m/s	**0.0000** m/s	Settles to full zero-velocity halt under active braking.
Target Standoff e_d	0.041 m	0.065 m	+**0.0200** m (2.00 cm)	Exact stop at $d = 0.220$ m (target: 0.200 m) via dead reckoning.
Heading Alignment e_θ	0.038 rad (2.1°)	0.082 rad (4.7°)	**0.015** rad (0.86°)	Visual servo centering keeps AprilTag securely inside optical FOV.

Key Analytical Comments:

- **Dynamic Lag vs. Steady-State Precision:** Longitudinal error $e_x(t)$ peaks at 0.751 m during mid-run cruise due to the chassis acceleration filter and motor inertial lag. However, during the terminal deceleration phase ($t > 10$ s), $e_x(t)$ rapidly contracts, yielding a steady-state terminal position error of just **1.96** cm.
- **Nonholonomic Invariance:** Lateral error $e_y(t)$ remains bounded below 2.8 cm throughout the run and finishes at 0.4 cm, validating that the commanded unicycle trajectory prevents chassis crabbing.
- **Pure-Vision Standoff Guard:** When the tag leaves the camera view at $d \leq 0.35$ m, incremental wheel integration tracks remaining distance. The active brake engages at $d = 0.200$ m, bringing the chassis to a complete stop at $d = 0.220$ m with zero external ToF sensor.